***Based on my coding work flow, here is a step-by-step guide of what I have just did:***

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## ➢ what is the tips data set:

Food servers' tips in restaurants may be influenced by many factors, including the nature of the restaurant, size of the party, and table locations in the restaurant. Restaurant managers need to know which factors matter when they assign tables to food servers. For the sake of staff morale, they usually want to avoid either the substance or the appearance of unfair treatment of the servers, for whom tips (at least in restaurants in the United States) are a major component of pay.
In one restaurant, a food server recorded the following data on all customers they served during an interval of two and a half months in early 1990. The restaurant, located in a suburban shopping mall, was part of a national chain and served a varied

menu. In observance of local law, the restaurant offered to seat in a non-smoking section to patrons who requested it. Each record includes a day and time, and taken together, they show the server's work schedule.

- **what graphs includes in the code**:
  1. line chart
  2. box plot
  3. heatmap
  4. kde plot
  5. ecdf plot
  6. violin plot
- what did I get from charts:
  - first observation is to get the most smoked day among them and found out the Friday was the most
  - bills for males increases day by day but for females it the lowest on Friday
  - smoker bills are the highest
  - tips increases as the size of the groups increases
  - males pay more tips than females
  - dinner tips are more than the lunch tips
  - there are some outliers that goes beyond 50$ on Saturday
  - the average of the bills paid is around 20$

- the difference between matplotlib and seaborn and an intro on plotly
    - **Matplotlib:** It is a Python library used for plotting graphs with the help of other libraries like Numpy and Pandas. It is a powerful tool for visualizing data in Python. It is used for creating statistical inferences and plotting 2D graphs of arrays. It was first introduced by John D. Hunter in 2002. It uses Pyplot to provide a MATLAB-like interface free and open-source. It is capable of dealing with various operating systems and their graphical backends.
    - **Seaborn:** It is also a Python library used for plotting graphs with the help of Matplotlib, Pandas, and Numpy. It is built on the roof of Matplotlib and is considered as a superset of the Matplotlib library. It helps in visualizing univariate and bivariate data. It uses beautiful themes for decorating Matplotlib graphics. It acts as an important tool in picturing Linear Regression Models. It serves in making graphs of statical Time-Series data. It eliminates the overlapping of graphs and also aids in their beautification.

| Features | Matplotlib | Seaborn |
|---|---|---|
| Functionality | It is utilized for making basic graphs. Datasets are visualized with the help of bar graphs, histograms, pie charts, scatter plots, lines, and so on. | Seaborn contains several patterns and plots for data visualization. It uses fascinating themes. It helps in compiling whole data into a single plot. It also provides the distribution of data. |
| Syntax | It uses comparatively complex and lengthy syntax. Example: Syntax for bar graph- matplotlib.pyplot.bar(x_axis, y_axis). | It uses comparatively simple syntax which is easier to learn and understand. Example: Syntax for bargraph- seaborn.barplot(x_axis, y_axis). |
| Dealing Multiple Figures | We can open and use multiple figures simultaneously. However, they are closed distinctly. Syntax to close one figure at a time: matplotlib.pyplot.close(). Syntax to close all the figures: matplotlib.pyplot.close("all") | Seaborn sets the time for the creation of each figure. However, it may lead to (OOM) out of memory issues |
| Visualization | Matplotlib is well connected with Numpy and Pandas and acts as a graphics package for data visualization in Python. Pyplot provides similar features and syntax as in MATLAB. Therefore, MATLAB users can easily study it. | Seaborn is more comfortable in handling Pandas data frames. It uses basic sets of methods to provide beautiful graphics in Python. |
| Pliability | Matplotlib is a highly customized and robust | Seaborn avoids overlapping plots with the help of its default themes |
| Data Frames and Arrays | Matplotlib works efficiently with data frames and arrays.It treats figures and axes as objects. It contains various stateful APIs for plotting. Therefore plot() like methods can work without parameters. | Seaborn is much more functional and organized than Matplotlib and treats the whole dataset as a single unit. Seaborn is not so stateful and therefore, parameters are required while calling methods like plot() |
| Use Cases | Matplotlib plots various graphs using Pandas and Numpy | Seaborn is the extended version of Matplotlib which uses Matplotlib along with Numpy and Pandas for plotting graphs |

## ❖ Plotly Open Source Graphing Library for Python

**Plotly** is a popular open-source Python library used for creating interactive, publication-quality visualizations. It is widely used in data science, analytics and machine learning for presenting data insights visually and interactively. It supports a wide variety of

charts including line plots, scatter plots, bar charts, pie charts, heatmaps and 3D plots. It integrates well with Jupyter notebooks, Dash and web applications.

***Important Facts to Know:***

- ***Interactive by Default:*** *Built-in support for zoom, pan, tooltips and legends enhances data exploration.*
- ***Web Integration:*** *Easily embeddable in web apps and dashboards using frameworks like Dash.*

***References :***


- ***https://www.kaggle.com/datasets/ranjeetjain3/seaborn-tips-dataset***
- ***. https://www.geeksforgeeks.org/python/difference-between-matplotlib-vs-seaborn/***
- ***https://www.geeksforgeeks.org/python/python-plotly-tutorial/***